MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6L7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pentagrid mixer

Outline	---	B-4 (EIA)
Base	---	B7-22
Envelope	---	MTT-8
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	7	8	Cap
Element	---	sh	h	a	g2,g4	g3	h	g5, k	g1

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec3	Ec2+4	Pp	Ehk	Pg2	Alt
Unit:	V	V dc	V dc	V dc	V dc	W	v	W	ft
Maximum:									
Amplifier:	6.9	330	---	---	110	1.5	100	1.0	See note 1
Mixer:	6.9	330	---	---	165	1.0	100	1.5	See note 1
Minimum:	5.7	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	250	-3	-3	100	---	---	---	---

GENERAL:

Qualification - Not required

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Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 1	2					
1266	Total grid current	3		Ic3	0	-2.0	µA dc
1256	Electrode current (anode)			Ib	3.6	7.0	mA dc
1231	Emission	3	Eb = Ec2 + 4 = Ec1 = Ec3 = 30 V dc	Is	45	---	mA dc
1256	Electrode current (screen)			Ic2+4	4.7	8.5	mA dc
1306	Transconductance		g1 to a	Sm	900	1,300	µmhos
1201	Short and discontinuity			---	---	---	---
	Quality conformance inspection, part 2						
1301	Heater current			If	275	325	mA
1336	Heater-cathode leakage			Ihk	---	20	µA dc
1261	Electrode voltage (grid 1)		Sm(g1/a) = 10.5 µmhos	Ec1	-20	-32	V dc
1261	Electrode voltage (grid 3)		Sm(g3/a) = 27.5 µmhos	Ec3	-9	-17	V dc
1031	Low-frequency vibration		Rp = 10,000 ohms	Ep	---	200	mV ac
1331	Direct-interelectrode capacitance	4		Cg1-all Cp-all Cg3-all Cgp	6.2 8.2 8.3 ---	8.8 13.8 11.7 0.0015	pF pF pF pF
1211	Insulation of electrodes	5		---	---	---	---
1101	Secureness of base, cap, or insert			---	---	---	---
1111	Base pin solder depth			---	---	---	---
1105	Permanence of marking			-	---	---	---
	Quality conformance inspection, part 3						
1501	Intermittent life		Group A; Ec2+4 = 150 V dc; Ec1 = Ec3 = -6 V dc; Ehk = 100 V	---	---	---	---
---	Intermittent life-test end point (500 hours)						
1306	Transconductance		g1 to a	Sm	800	-	µmhos

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NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. The AQL (percent defective) for each test listed under QCI, part 1, shall be 0.65, inspection level II.
3. This test shall be performed at the conclusion of the holding period.
4. The AQL (percent defective) shall be 6.5, inspection code E.
5. The AQL (percent defective) shall be 4.0, inspection level S3.
6. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Air Force - 11, 99
DLA - ES

User activities:
Army - ME
Navy - AS, OS, MC, CG
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Army - ER

(Project 5960-3326)

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